

Ivan Romero

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Software Engineer with 3 years of experience building backend systems and microservices in Python. Expertise in Django, REST APIs, Docker, and distributed system design. Proven experience leading engineering teams and deploying production solutions for major Spanish banking clients processing high-volume document workflows.

Skills

Programming & Tools: Python, SQL, Bash, JavaScript, Git, PyTest, Prometheus

ML & AI: PyTorch, CNNs, NLP/NER, LLMs, NumPy, Pandas, Agentic AI, Object Detection

Backend & Infra: FastAPI, Django, PostgreSQL, Redis, Docker, Nginx, CI/CD, AWS EC2, AWS S3, Hetzner

Languages: English (fluent), Spanish (native), Catalan (native)

Work experience

Software Engineer | Serimag | Barcelona, Spain **Feb 2024 – Aug 2025**

- Mentored and led a team of 3 engineers on Django/FastAPI/REST API development, delivering 5+ production microservices for a major Spanish banking client.
- Evolved and optimized core Python orchestration platform coordinating business logic across 10+ microservices (including data-intensive ML pipelines) reducing API payload and latency for 100,000+ daily documents.
- Led the design and implementation of a High Availability solution for OAuth2.0 services, deploying MySQL InnnoDB Cluster with load balancing and automatic failover across dual environments, reducing recovery time from hours to under one minute.
- Owned the end-to-end deployment lifecycle with weekly microservice releases on Linux using Docker, Nginx, CI/CD pipelines, automated testing with PyTest, and production monitoring.

AI Engineer Intern | Serimag | Barcelona, Spain **Jan 2023 – Feb 2024**

- Led the development of real-time CNN-based classification models processing 30,000+ documents daily for a major Spanish bank, reducing manual workload by 80%.
- Engineered NLP/NER pipelines that increased invoice information extraction automation from 60% to 70% at 99% accuracy, saving 35 manual hours weekly across 5,000+ daily documents.
- Audited and improved production model, maintaining misclassification rate under 1% in live banking workflows.

Full Stack Developer | Bandalux | Barcelona, Spain **June 2021 – Sept 2021**

- Implemented backend features using .NET and enhanced user interfaces through JavaScript, HTML and CSS for internal web applications, improving usability and performance.

Education

Polytechnic University of Catalonia – Barcelona, Spain **Sept 2024 – May 2026 (expected)**
Master's in Advanced Telecommunications Technologies (Specialization in Deep Learning for Multimedia Processing)

Polytechnic University of Catalonia – Barcelona, Spain **Sept 2018 – Jan 2024**
Bachelor's in Telecommunications Engineering

Projects

Business Intelligence Platform | Client: Rivex **Sept 2025 – Jan 2026**

- Architected and developed a BI web app for Barcelona's White Rabbit Museum to visualize marketing ROI and customer trends using FastAPI, PostgreSQL, Redis and React, deploying containerized microservices on AWS EC2 via GitHub Actions CI/CD pipeline with Nginx reverse proxy and Prometheus/Grafana observability.

4Forms – Digitalization Platform | Client: Engi-On **Sept 2025 – Dec 2025**

- Built a Django REST Framework/React platform on AWS EC2 that eliminated 100% of manual paper-based workflows for industry machinery maintenance; automated CI/CD deployment and Prometheus/Grafana monitoring.